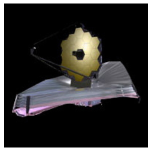

9447 - GRB 250702B/D/E Burning Red: A GRB or TDE?

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1		NIRCam Imaging	(1) GRB250702B

ABSTRACT

GRB 250702B/D/E is one of the longest "gamma-ray bursts" (GRBs) ever detected with a T90 of over 15,000s, earning it a designation in the 'ultralong' GRB subclass. The discovery of a fast-fading, very red (H-K = 2.5 mag) NIR counterpart localized to an edge-on galaxy strongly supports its classification as an extragalactic transient. The extraordinary duration of GRB 250702B/D/E, and the discovery of soft X-ray emission discovered ~24 hrs prior to the gamma-ray trigger, however, brings into question its designation as a true "gamma-ray burst." An alternative identity for the source is as a tidal-disruption event (TDE). The multi-wavelength properties, including the precursor X-ray emission, are consistent with a jetted tidal-disruption event, however the off-nuclear position of GRB 250702B/D/E would make this the first jetted TDE of its kind. Only 3 other off-nuclear TDEs have been discovered and are thought to be powered by massive black holes. TDEs involving intermediate-mass black holes (IMBH) are also expected to be found in off-nuclear locations. We propose to use JWST/NIRCam to precisely measure the SED and time evolution of GRB 250702B/D/E, further resolve the field around the candidate NIR counterpart, and identify the progenitor of this extraordinary event.

OBSERVING DESCRIPTION

We are proposing to precisely measure the SED and time evolution of GRB 250702B/D/E. We request one epoch of NIRCam imaging in each F150W/F277W, F150W/F356W, and F200W/F444W. The SW filter F150W is repeated so as to reach a high enough SNR. We request the B module, FULL array, SHALLOW4 readout pattern, with 8 groups per integrations, and 3 integrations per exposure. We request specific PAs so as to avoid diffraction spikes from nearby stars.

Proposal 9447 - Targets - GRB 250702B/D/E Burning Red: A GRB or TDE?

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	GRB250702B	RA: 18 58 45.5700 (284.6898750d) Dec: -07 52 26.20 (-7.87394d) Equinox: J2000		

Comments:
Category=Star
Description=[Gamma Ray bursters]

Observation

Proposal 9447, Observation 1

Wed Aug 27 15:01:02 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	GRB250702B	RA: 18 58 45.5700 (284.6898750d) Dec: -07 52 26.20 (-7.87394d) Equinox: J2000		

Comments:
Category=Star
Description=[Gamma Ray bursters]

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		STANDARD		4

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W	F277W	SHALLOW4	8	3	12	4	5110.703	
2	F150W	F356W	SHALLOW4	8	3	12	4	5110.703	
3	F200W	F444W	SHALLOW4	8	3	12	4	5110.703	

Special Requirements

Aperture PA Range 25.05262691 to 65.05262691 Degrees (V3 25.0 to 65.0)
Aperture PA Range 85.05262691 to 125.05262691 Degrees (V3 85.0 to 125.0)
Aperture PA Range 145.05262691 to 185.05262691 Degrees (V3 145.0 to 185.0)
Aperture PA Range 205.05262691 to 245.05262691 Degrees (V3 205.0 to 245.0)
Aperture PA Range 265.05262691 to 305.05262691 Degrees (V3 265.0 to 305.0)
Aperture PA Range 325.05262691 to 5.05262691 Degrees (V3 325.0 to 5.0)
Offset 38.0 arcsec, 38.0 arcsec